PRICE $2.00

HEATH COMPANY • BENTON HARBOR, MICHIGAN

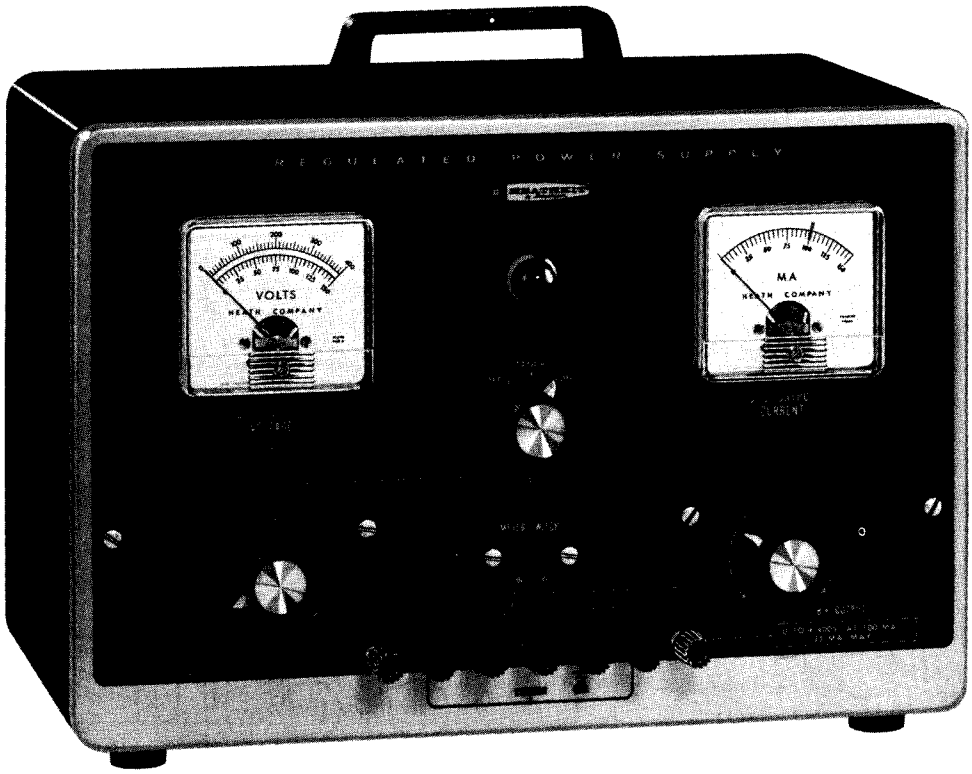


HEATHKIT®
ASSEMBLY MANUAL

REGULATED POWER SUPPLY
MODEL IP-32

ASSEMBLY AND OPERATION OF THE HEATHKIT REGULATED POWER SUPPLY

MODEL IP-32

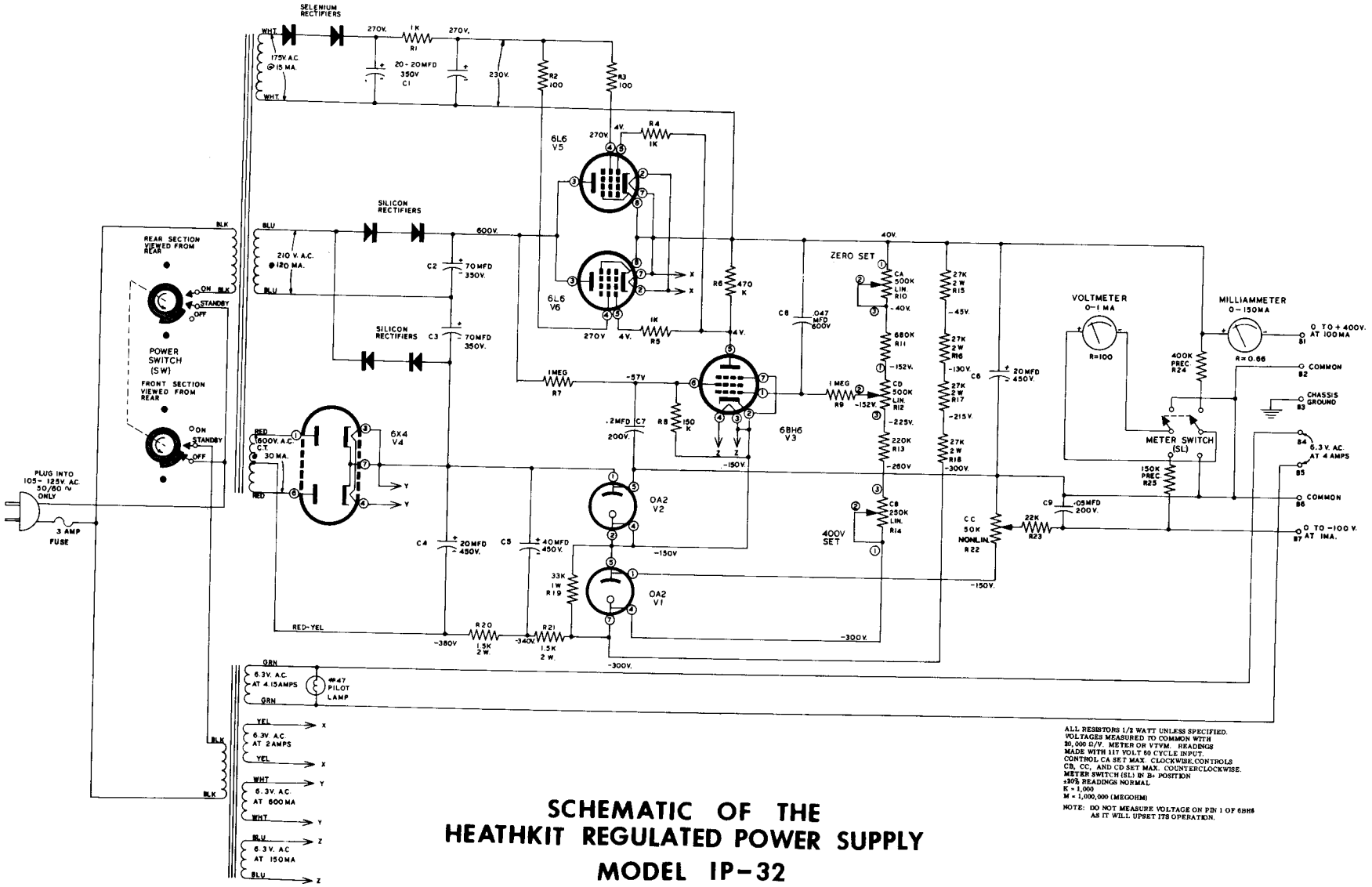


SPECIFICATIONS

Power Requirements:	105 to 125 volts AC, 50/60 cycle, 150 watts maximum
Output:	
B+ Voltage:	0 to 400 volts DC regulated; 0 to 100 ma continuous (125 ma intermittent)
Bias Voltage:	0 to -100 volts DC at 1 ma
Filament Voltage:	6.3 volts AC at 4 amps (Insulated to withstand 1500 volts DC)
B+ Regulation:	Output variation less than 1% from no load to full load, for outputs of 100 to 400 volts. Output variation less than ±0.5 volt for a ±10 volt variation in the 117 volt AC input
B+ Ripple:	Less than 10 mv RMS ripple, jitter and noise
B+ Output Impedance:	Less than 10 Ω from DC to 1 mc (See output impedance curve.)

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Heath Company

10/19/62



SCHEMATIC OF THE HEATHKIT REGULATED POWER SUPPLY MODEL IP-32

ALL RESISTORS 1/2 WATT UNLESS SPECIFIED.
VOLTAGES MEASURED TO COMMON WITH 20,000 Ω/V. METER OR VTVM. READINGS MADE WITH 117 VOLT 60 CYCLE INPUT.
CONTROL CA SET MAX. CLOCKWISE, CONTROLS CB, CC, AND CD SET MAX. COUNTERCLOCKWISE.
METER SWITCH (SL) IN B+ POSITION
±20% READINGS NORMAL
K = 1,000
M = 1,000,000 (MEGOHM)

NOTE: DO NOT MEASURE VOLTAGE ON PIN 1 OF 6BH6 AS IT WILL UPSET ITS OPERATION.

HEATHKIT by Daystrom

Meters:
Voltmeter:..... 0 to 400 volts or 0 to 150 volts
Milliammeter:..... 0 to 150 ma

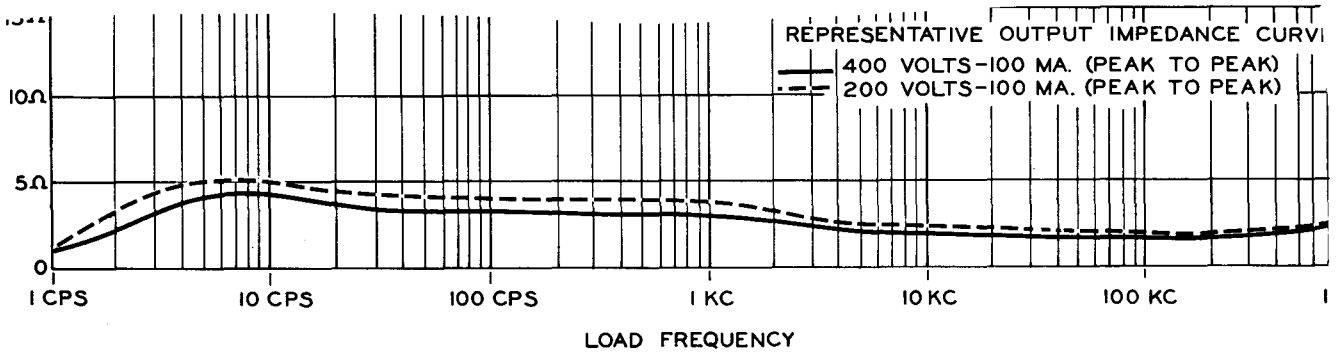
Tube and Diode Complement:..... 2 - 6L6 Series Regulators
1 - 6BH6 Control Amplifier
1 - 6X4 Bias Rectifier
2 - OA2 Bias Voltage Regulators
2 - 50 ma Selenium Screen Voltage Rectifiers
4 - 500 ma Silicon High Voltage Rectifiers

Output Terminals provided:..... B+ and common
C- and common
6.3 VAC filament
Chassis ground

Dimensions:..... 13" wide x 8 1/2" high x 7" deep

Net Weight:..... 12 pounds

Shipping Weight:..... 16 pounds



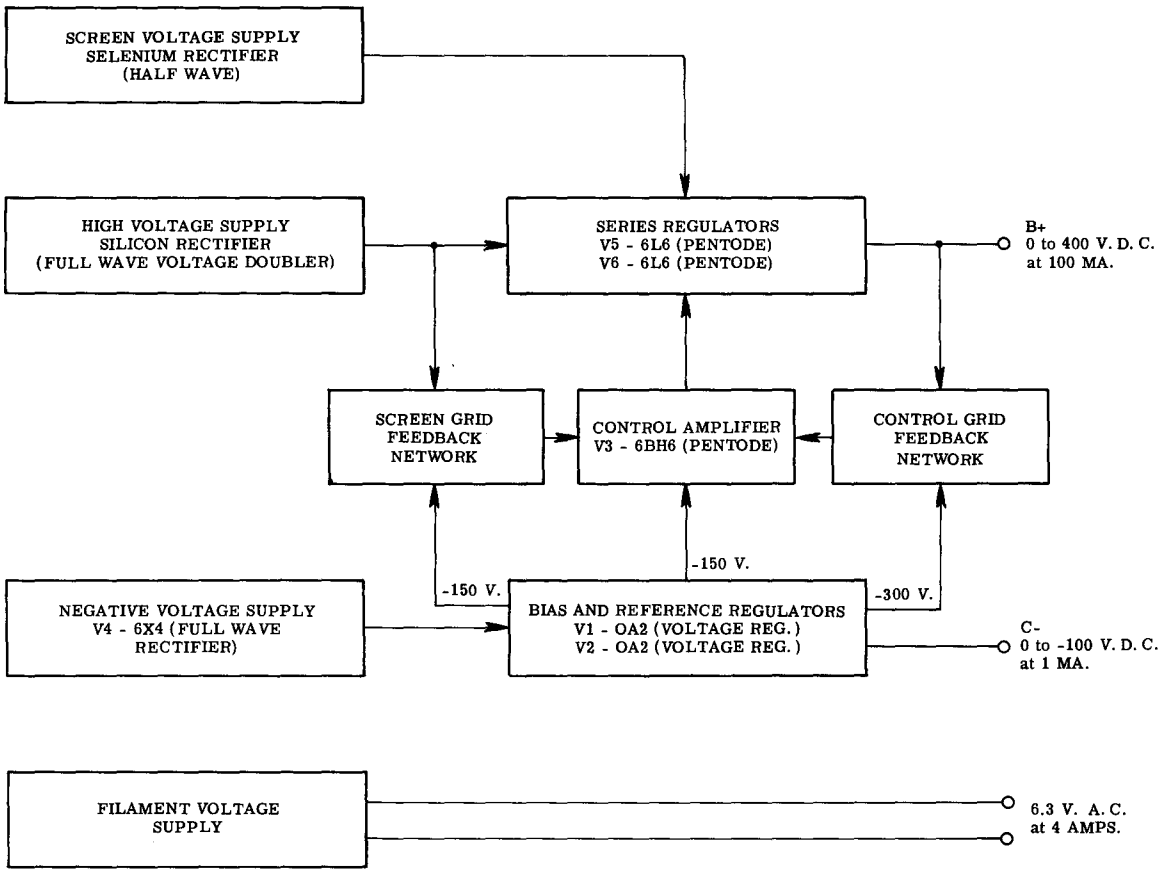
INTRODUCTION

The Heathkit Regulated Power Supply, Model IP-32, has been designed as a convenient source of variable regulated B+ voltage, variable bias voltage, and filament voltage for laboratories and workshops. This supply allows the designer, or experimenter, to develop and test his circuitry without having to construct a power supply each time. The output voltage and current are both continuously monitored by separate panel meters, allowing easy determination of power supply requirements for a particular piece of equipment.

The outputs are isolated from the chassis to allow the B+ to be used as either a positive or negative supply. Separate filament and high voltage transformers are employed so that all high voltage can be switched off, leaving only filament voltage available while changes in circuitry are made. This eliminates filament warmup time.

The bias voltage output control has a special taper for fine adjustment at low voltages and a meter switch is provided to read either the bias or B+ voltage. Built-in circuit protection prevents damage to the unit if the bias voltage output should accidentally be shorted. The supply is also fused for protection against overloads and short circuits.

CIRCUIT DESCRIPTION



The Heathkit Regulated Power Supply is, in essence, a power supply in which the output current may vary over a wide range (from 0 to 100 ma, depending on the load), while the output voltage remains constant at its original setting.

The high voltage (B+) is supplied from a full wave voltage doubler, using silicon diode rectifiers. This voltage is connected to the plates of the two paralleled 6L6 series regulator tubes, the cathodes of which are connected to the output through the milliammeter. To allow the 6L6's to operate as pentodes*, a separate power supply is employed to supply the screen grid voltage. This supply consists of a half-wave rectifier, using a pair of selenium diode rectifiers feeding a dual section capacity filter. A pair of 100Ω resistors (parasitic suppressors) couples the supply voltage to the screen grids. The common for this supply is returned to the 6L6 cathodes, thus maintaining a relatively constant voltage on the 6L6 screen grids.

Operating in this configuration, the paralleled 6L6's act as a large variable resistor, the value of which is controlled by a small voltage applied to the control grids. This voltage is produced by the 6BH6 control tube, which operates as a DC voltage feedback amplifier. Its grid is fed from a DC voltage divider connected to the B+ output, while its plate is direct coupled to the 6L6 grids by a pair of 1 KΩ (current limiting parasitic suppressor) resistors. In this manner, any voltage variation** at the B+ output is immediately amplified and fed back (with reverse polarity) to the grids of the paralleled 6L6 series regulator tubes. This produces a corresponding resistance change in the series regulators which is in opposition to the output variation, thus cancelling it out.

*Pentode operation is superior to triode operation in most wide range series regulator applications.
**Voltage variations - caused by changes in output load and/or input voltage.

Screen voltage for the control amplifier is taken from a DC voltage divider, connected between the high voltage supply and the regulated -150 volt supply. The high voltage varies inversely to changes in the output current, and this variation is fed into the screen grid through the divider network. The output impedance of the Power Supply can be made positive or negative by using different values in this divider. Actual values for this network are chosen to give substantially zero output impedance.

The negative output voltage is produced by a 6X4 full wave rectifier, feeding a two section capacity filter. The output of this filter feeds a pair of OA2 voltage regulators connected in series, to supply a stable -150 and -300 volts. These two voltages supply a reference for the 6BH6 control amplifier and its DC divider networks.

The bleeder resistor network, consisting of four 27 KΩ resistors in series, is connected between the B+ output and the -300 volt source. In this manner, current flow through the series regulators is maintained at all output voltage settings.

The 50 KΩ control, connected across the -150 volt source, provides the variable bias voltage, which is coupled to its output terminal through a 22 KΩ (current limiting) resistor. This resistor prevents damage to the unit in event the bias output should accidentally be shorted.

Heater voltage for the unit is supplied by a separate transformer. This transformer also supplies 6.3 volts at 4 amperes for the external filament voltage output.

PRELIMINARY CONSTRUCTION NOTES

UNPACK THE KIT CAREFULLY AND CHECK EACH PART AGAINST THE PARTS LIST. If some shortage or parts damage is found in checking, please notify us promptly and include all inspection slips with your letter.

The large, fold-in pictorials are reproductions of pictorials appearing in the manual. They have been prepared in this manner to be attached to the wall above your work space, to help in the construction of your instrument.

The following instructions are presented in a logical, step-by-step sequence to enable you to complete your kit with the least possible confusion. Be sure to read each step all the way through before you start to do it. When the step is completed, check it off in the space provided. This is particularly important as it may prevent omissions or errors, especially if your work is interrupted.

MOUNTING PARTS

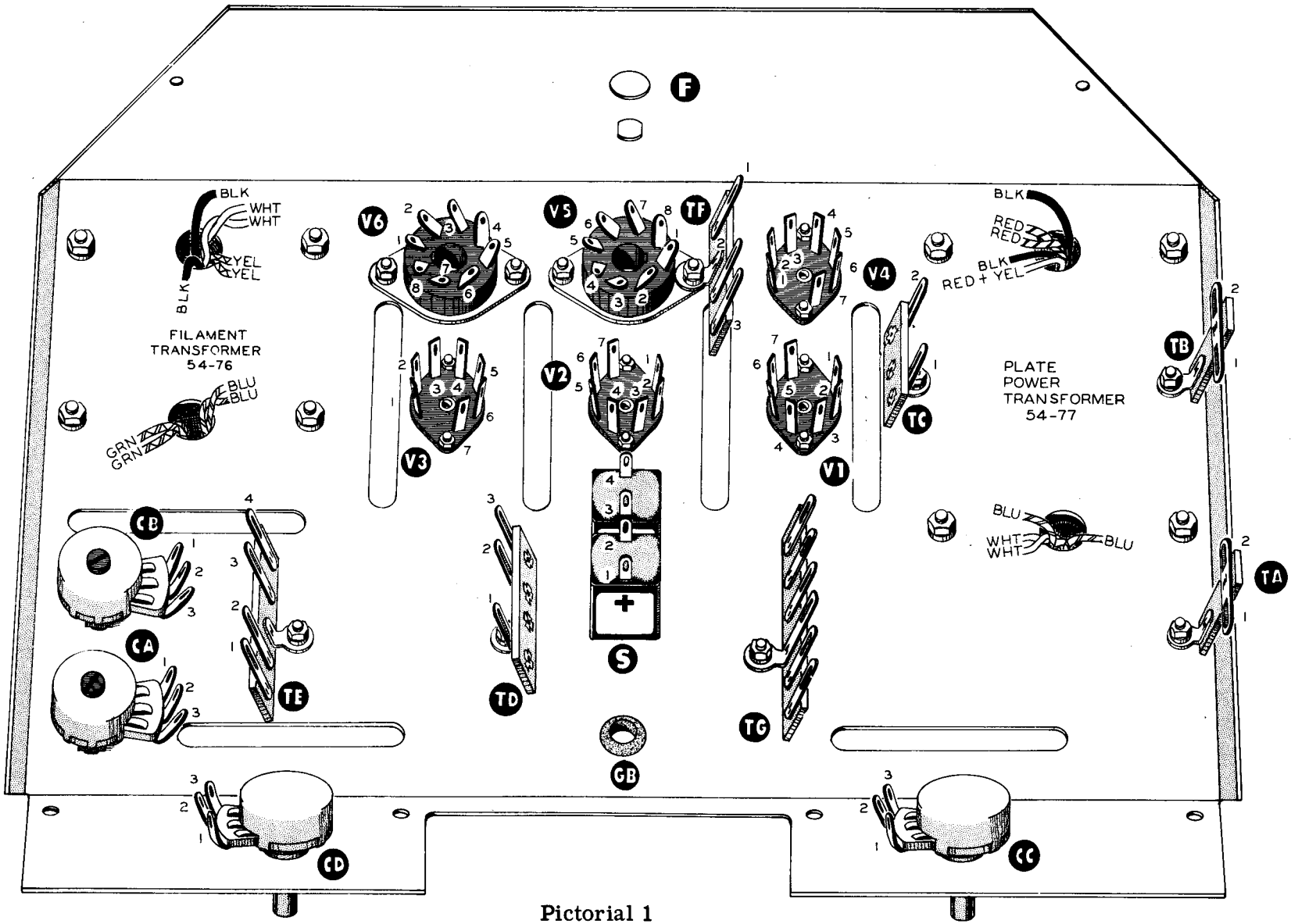
( ) Locate the chassis and orient upside down as shown in Pictorial 1 on Page 6.

( ) Mount a 1/2" grommet in hole GB.

( ) Mount 7-pin tube sockets in locations V1, V2, V3 and V4. Orient with blank spaces positioned as shown in Pictorial 1, and secure with 3-48 x 1/4" binder head machine screws and nuts.

( ) Mount octal tube sockets (8-pin) in locations V5 and V6. Orient with keyway positioned as shown in Pictorial 1, and secure with 6-32 x 3/8" binder head machine screws, lockwashers and nuts. Mount a 3-lug terminal strip under one lockwasher and nut on socket V5 in location TF. Note that there are two types of 3-lug terminal strips furnished in the kit. Be sure to use the correct one and orient as shown.

( ) Mount 1-lug terminal strips in locations TA and TB. Orient as shown in Pictorial 1, and secure with 6-32 x 3/8" binder head machine screws, lockwashers and nuts.


F
BLK
WHT
WHT
YEL
YEL
BLK
FILAMENT
TRANSFORMER
54-76
BLU
BLU
GRN
GRN
V6
V5
TF
V4
V2
V3
V1
TC
BLK
RED
RED
BLK
RED + YEL
PLATE
POWER
TRANSFORMER
54-77
TB
TA
BLU
WHT
WHT
BLU
CB
CA
TE
TD
S
TG
GB
CD
CC

Pictorial 1

( ) Mount a 2-lug terminal strip in location TC. Orient as shown, and secure with 6-32 x 3/8" binder head machine screw, lockwasher and nut.

( ) Mount a 3-lug terminal strip in location TD. Orient as shown, and secure with 6-32 x 3/8" binder head machine screw, lockwasher and nut.

( ) Mount a 4-lug terminal strip in location TE. Orient as shown, and secure with 6-32 x 3/8" binder head machine screw, lockwasher and nut.

( ) Mount the two silicon rectifier holders in location W, placing 6-32 x 3/8" round head machine screws through the holder and chassis, and securing on the top of the chassis with lockwashers and nuts. Be sure to orient as shown in Pictorial 1.

( ) Mount a 6-lug terminal strip at location TG. Orient as shown, and secure with a 6-32 x 3/8" binder head machine screw, lockwasher and nut.

( ) Mount a 500 KΩ control (#10-60) in location CA, and 250 KΩ control (#10-59) in location CB. Orient the controls as shown in Pictorial 1, inserting the tabs through the holes in the chassis and twisting them to secure. See Figure 1.

( ) Mount the plate power transformer (#54-77) in its proper location as shown in Pictorial 1. Orient with black and red leads coming through the hole nearest the rear apron. Secure with four 8-32 x 3/8" round head machine screws, #8 flat washers, lockwashers, and nuts. The flat washers should be placed under the heads of the mounting screws.

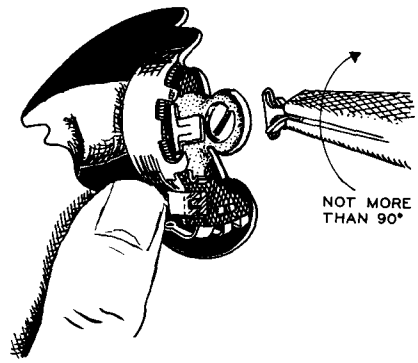


Figure 1

( ) NOTE: The plate power transformer (#54-77) leads have been pre-cut to length. However, the filament transformer (#54-76) is used in several kits, which makes pre-cut leads impractical. Therefore, before mounting the filament transformer, cut its leads to the following lengths and strip and tin the ends of each lead.

( ) White - 8" ( ) Blue - 4" ( ) Yellow - 3" ( ) Green - 11-1/2" ( ) Black - 11"
( ) White - 8" ( ) Blue - 4" ( ) Yellow - 3" ( ) Green - 11-1/2" ( ) Black - 7"

( ) Mount the filament transformer (#54-76) in its proper location as shown in Pictorial 1. Orient with yellow, white and black leads coming through the hole nearest the rear apron. Secure with four 8-32 x 3/8" round head machine screws, #8 flat washers, lockwashers, and nuts. The flat washers should be placed under the heads of the mounting screws.

( ) Temporarily mount a 50 KΩ control (#10-89) in location CC. Orient as shown in Pictorial 1 and secure with lockwasher, flat washer and nut as shown in Figure 2.

( ) Temporarily mount a 500 KΩ control (#10-26) in location CD. Orient as shown in Pictorial 1 and secure with a lockwasher, flat washer and nut as shown in Figure 2.

SWITCH & CONTROL MOUNTING DETAIL

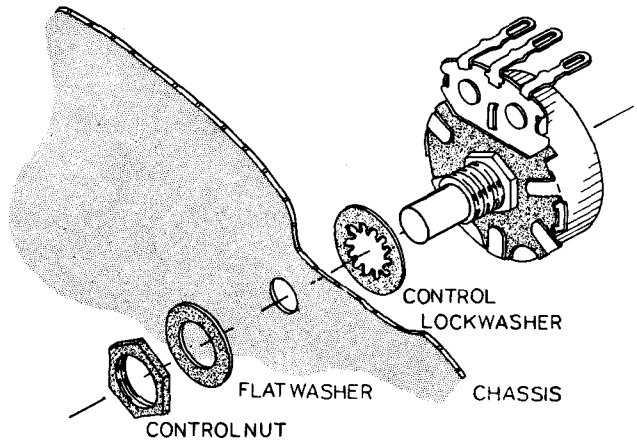


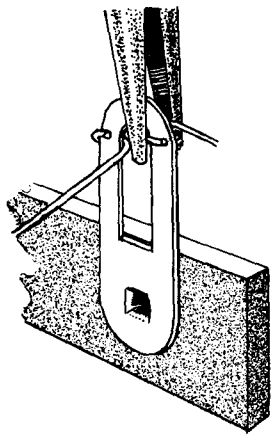
Figure 2

PROPER SOLDERING TECHNIQUES

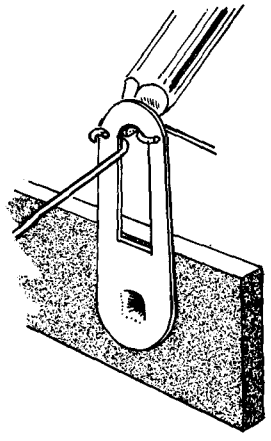
Only a small percentage of Heathkit purchasers find it necessary to return an instrument for factory service. Of these instruments, by far the largest proportion malfunction due to poor or improper soldering.

Correctly soldered connections are essential, if the performance engineered into a kit is to be fully realized. If you are a beginner with no experience in soldering, a half hour's practice with some odd lengths of wire may be a worthwhile investment.

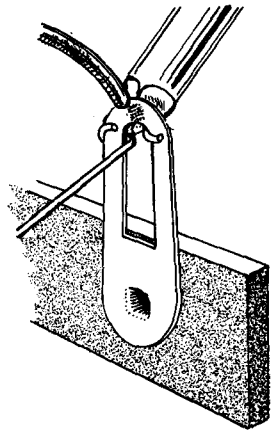
For most wiring, a 30 to 100 watt iron, or the equivalent in a soldering gun, is very satisfactory. Smaller irons generally will not heat the connections enough to flow the solder smoothly over the joint. Keep the iron tip clean and bright by wiping it from time to time with a rag.



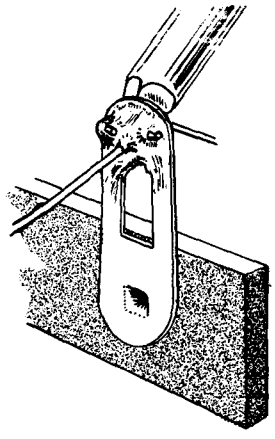
CRIMP WIRES



HEAT CONNECTION



APPLY SOLDER



ALLOW SOLDER TO FLOW

SOLDERING PROCEDURE

1. If the connection is to be made with hookup wire, strip 1/4" insulation from the end to be connected.

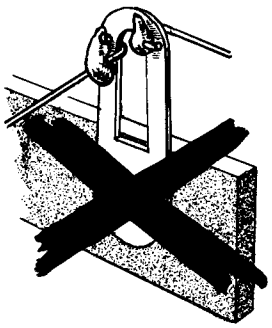
2. Crimp or bend the lead (not more than once) around the terminal, to form a good joint without relying on solder for physical strength.

3. Position the work, if possible, so that gravity will help to keep the solder where you want it.

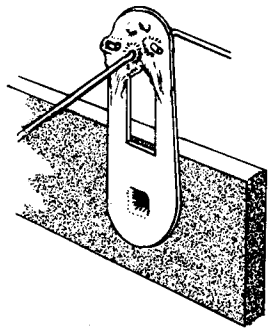
4. Place the tip of the soldering iron against the joint to be soldered until it is heated sufficiently to melt solder.

5. Then place the solder against the terminal and it will immediately flow out over the joint. Use only enough solder to thoroughly wet the junction; it is not necessary to fill the entire hole in the terminal with solder.

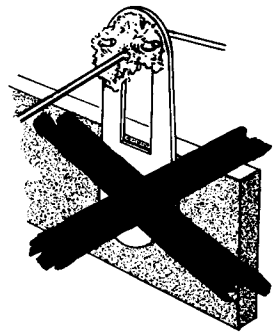
6. Remove the solder, and then the iron from the completed junction. Use care not to move the leads until the solder has solidified.



COLD SOLDER JOINT
CONNECTION INSUFFICIENTLY
HEATED



PROPER SOLDER
CONNECTION



COLD SOLDER JOINT
CONNECTION MOVED
WHILE COOLING

A poor or cold solder joint will usually look crystalline and have a grainy texture, or will stand up in a blob and not have adhered to the joint at all. Such joints should be reheated until the solder flows smoothly over the entire junction. In some cases it may be necessary to add a little more solder to achieve a smooth, bright appearance.

ROSIN CORE SOLDER HAS BEEN SUPPLIED WITH THIS KIT. THIS TYPE OF SOLDER MUST BE USED FOR ALL SOLDERING IN THIS KIT. ALL GUARANTEES ARE VOIDED AND WE WILL NOT REPAIR OR SERVICE EQUIPMENT IN WHICH ACID CORE SOLDER OR PASTE FLUXES HAVE BEEN USED. IF ADDITIONAL SOLDER IS NEEDED, BE SURE TO PURCHASE ROSIN CORE (60:40 or 50:50 TIN-LEAD CONTENT) RADIO TYPE SOLDER.

WIRING NOTES

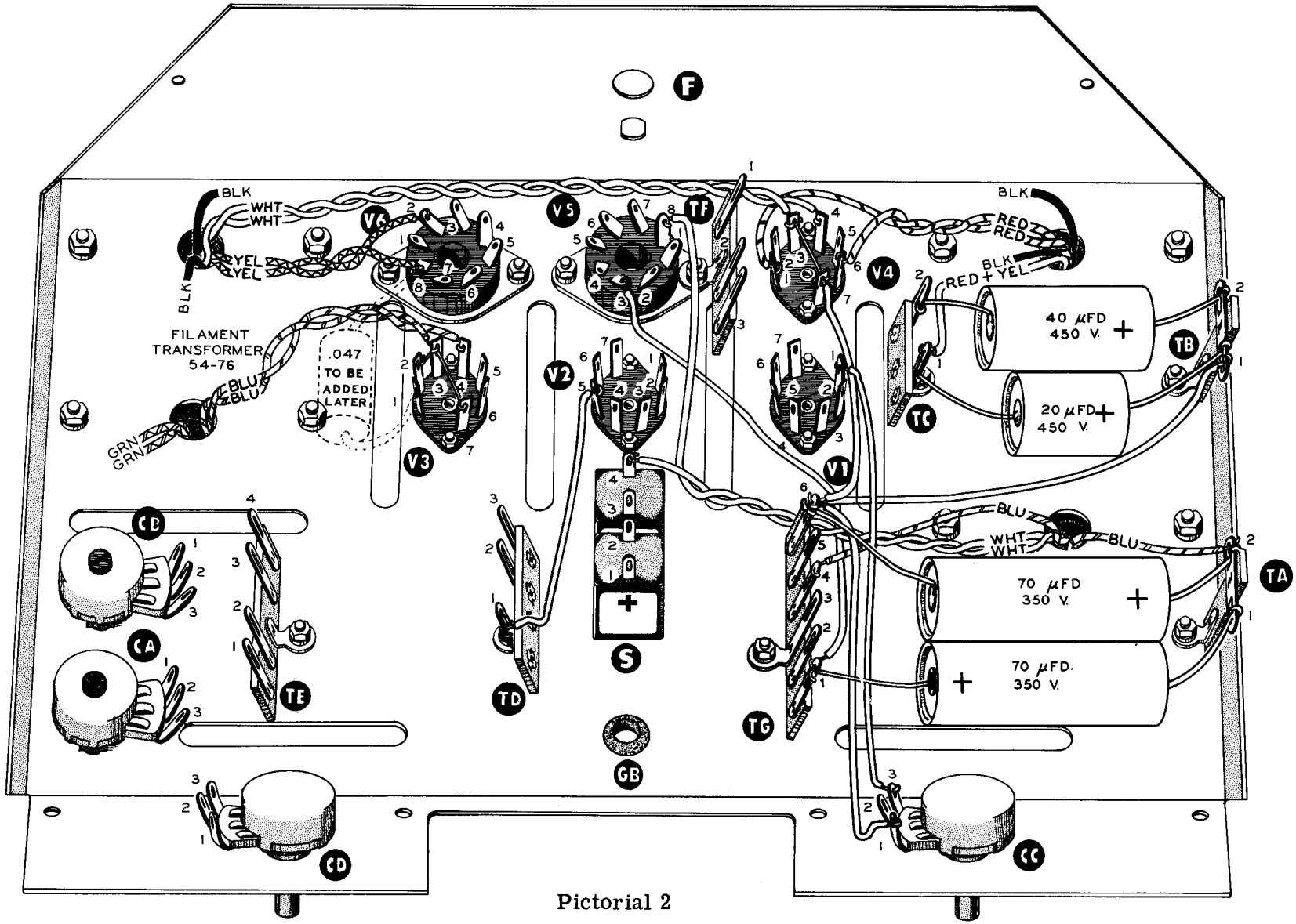
1. Unless otherwise indicated, all wire used is insulated (hookup wire).

2. Wherever there is a possibility of bare leads shorting to other bare leads or to the chassis, the leads should be covered with insulating sleeving. This is indicated in the instructions by the phrase "use sleeving."

3. Leads on resistors, capacitors and transformers are generally much longer than they need to be to make the indicated connections. In these cases, cut the excess lead off when connecting the part.

WIRING

NOTE: Steps in the Step-By-Step are presented simply and briefly. For example, if the step reads: "Connect a piece of hookup wire from TA2 (NS) to TD3 (S-3)," it will be understood that one end of a lead is to be connected to lug 2 of terminal strip TA and the other end to lug 3 of terminal strip TD. The (NS) means "no solder" as more leads will be connected later. The (S-3) means "solder this connection"; it should have three leads connected to it.


F
BLK
WHT
WHT
YEL
YEL
BLK
V6
V5
TF
BLK
RED
RED
BLK
RED + YEL
V4
40 μFD
450 V. +
TB
FILAMENT
TRANSFORMER
54-76
.047
TO BE
ADDED
LATER
BLU
BLU
GRN
GRN
V2
20 μFD +
450 V.
TC
V3
V1
CB
BLU
WHT
WHT
BLU
TA
70 μFD
350 V. +
CA
TE
TD
S
70 μFD.
350 V.
TG
GB
CD
CC

Pictorial 2

( ) Connect a piece of hookup wire from TG1 (NS) to socket V5, pin 3 (NS).

NOTE: When wiring to tube socket pins with two holes in them, always use the top holes (farthest from the chassis), unless instructed otherwise.

( ) Connect the longest blue lead from the power transformer (#54-77) to TG4 (NS).

( ) Connect the other blue lead from the power transformer to TA2 (NS).

( ) Connect the positive (+) lead of a 70 μfd 350 V electrolytic capacitor to TG1 (NS). Connect the negative (-) lead to TA1 (S-1).

( ) Similarly, connect the positive (+) lead of a 70 μfd 350 V electrolytic capacitor to TA2 (S-2). Connect the negative (-) lead to TG6 (NS).

( ) Twist together the two red leads from the power transformer. Connect one lead to V4, pin 1 (S-1) and the other lead to V4, pin 6 (S-1).

( ) Connect the red-yellow lead from the power transformer to TC1 (NS).

( ) Twist together the two white leads from the power transformer. Connect the longest lead to V5, pin 8 (NS) and the shortest lead to S4 (S-1).

( ) Connect a piece of bare wire from V4, pin 3 (NS) to V4, pin 7 (NS).

( ) Connect a piece of bare wire from V3, pin 7 (NS) through V3, pin 3 (NS) to V3, pin 2 (S-1).

( ) Twist together the two white leads from the filament transformer (#54-76). Connect one lead to V4, pin 3 (S-2) and the other to V4, pin 4 (S-1).

( ) Twist together the two yellow leads from the filament transformer. Connect one lead to V6, pin 2 (NS) and the other lead to V6, pin 8 (NS).

( ) Twist together the two blue leads from the filament transformer. Connect one lead to V3, pin 3 (S-2) and the other lead to V3, pin 4 (S-1). Run these leads close to socket V6 to insure room for .047 μfd capacitor, which will be added later.

( ) Connect the positive (+) lead of a 40 μfd 450 V electrolytic capacitor to TB2 (S-1). Connect the negative (-) lead to TC2 (NS).

( ) Connect the positive (+) lead of a 20 μfd 450 V electrolytic capacitor to TB1 (NS). Connect the negative (-) lead to TC1 (NS).

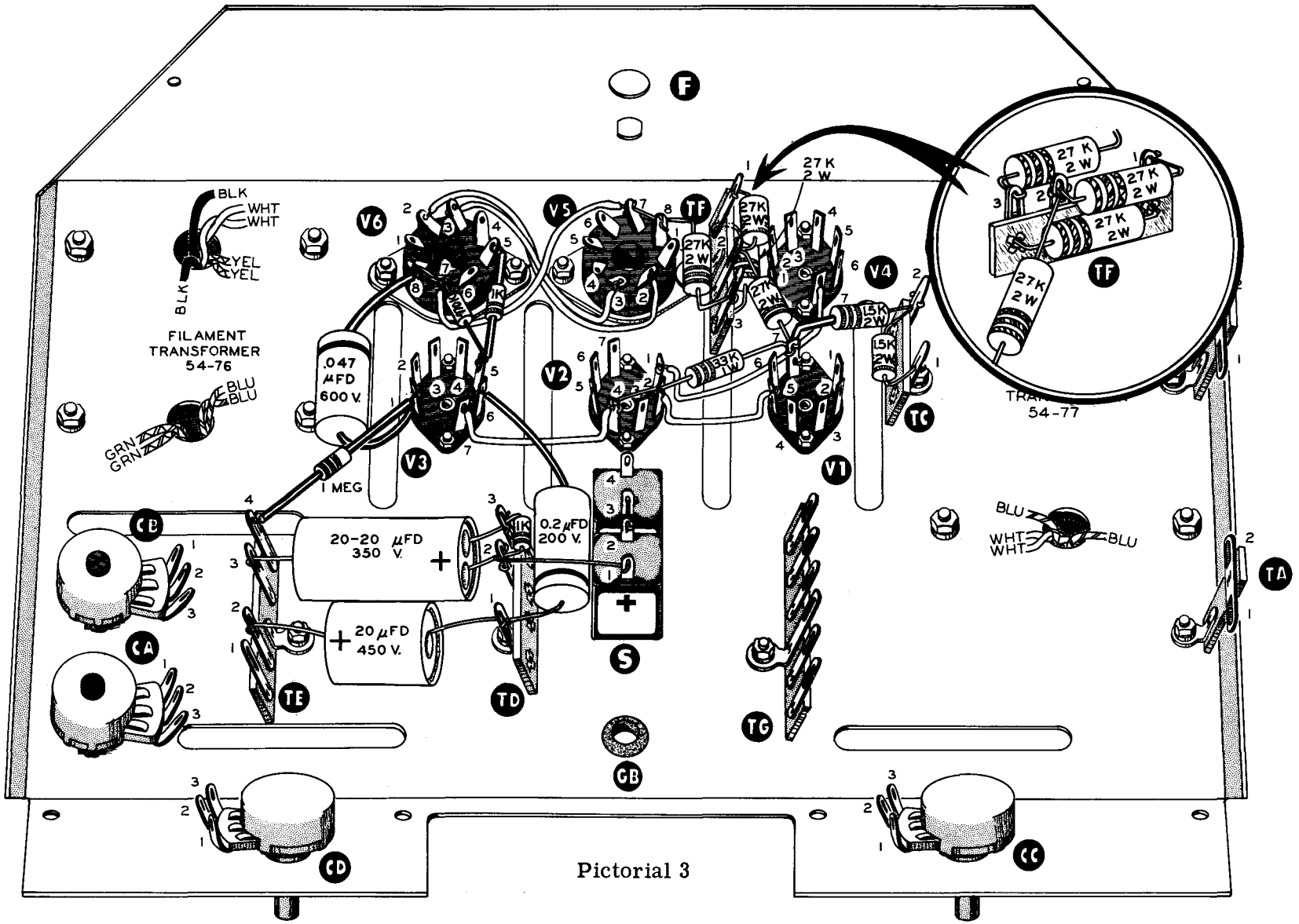
( ) Connect a piece of hookup wire from TB1 (S-2) to TG6 (NS).

( ) Connect a piece of hookup wire from V4, pin 7 (NS) to TG6 (NS).

( ) Connect a piece of hookup wire from V2, pin 5 (S-1) to TD1 (NS).

( ) Connect a piece of hookup wire from CC3 (S-1) to V1, pin 1 (S-1).

( ) Connect a piece of hookup wire from CC1 (NS) to TG6 (NS).


F
BLK
WHT
WHT
YEL
YEL
BLK
FILAMENT
TRANSFORMER
54-76
BLU
BLU
GRN
GRN
V6
V5
TF
27K
2W
27K
2W
V4
1.5K
2W
33K
1W
V2
.047
μFD
600 V.
1 MEG
V3
V1
TC
TRA
54-77
CB
20-20 μFD
350 V.
0.2 μFD
200 V.
20 μFD
450 V.
S
TE
TD
TG
BLU
WHT
WHT
BLU
TA
CA
GB
CD
CC

Pictorial 3

( ) Connect a piece of hookup wire from V4, pin 7 (S-3) to V2, pin 1 (S-1). See Pictorial 3.

( ) Connect a piece of hookup wire from V1, pin 5 (S-1) to V2, pin 2 (NS).

( ) Connect a 1.5 KΩ 2 watt resistor (brown-green-red) from TC1 (S-3) to TC2 (NS).

( ) Connect a 1.5 KΩ 2 watt resistor (brown-green-red) from TC2 (S-3) to V1, pin 7 (NS). (Use the bottom hole in the tube socket lug.) Use sleeving.

( ) Connect a piece of hookup wire from V5, pin 2 (S-1) to V6, pin 2 (S-2).

( ) Connect a piece of hookup wire from V5, pin 3 (NS) to V6, pin 3 (S-1).

( ) Connect a piece of bare wire from V5, pin 7 (NS) to V5, pin 8 (NS).

( ) Connect a piece of bare wire from V6, pin 7 (NS) to V6, pin 8 (NS).

( ) Connect a piece of hookup wire from V5, pin 7 (S-2) to V6, pin 7 (NS).

( ) Connect a 27 KΩ 2 watt resistor (red-violet-orange) from the bottom eyelet of lug TF1 to the bottom eyelet of lug TF3. Solder both bottom holes.

( ) Connect a 27 KΩ 2 watt resistor (red-violet-orange) from TF1 (S-1) to TF2 (NS).

( ) Connect a 27 KΩ 2 watt resistor (red-violet-orange) from TF2 (S-2) to V1, pin 7 (NS).

( ) Connect a 27 KΩ 2 watt resistor (red-violet-orange) from TF3 (S-1) to V5, pin 8 (S-3).

( ) Connect a 33 KΩ 1 watt resistor (orange-orange-orange) from V1, pin 7 (S-3) through V2, pin 2 (S-2) to V2, pin 4 (NS). Be sure to use the top holes in the tube socket pins.

( ) Connect a piece of hookup wire from V2, pin 4 (NS) to V3, pin 7 (S-2).

( ) Connect a 0.2 μfd 200 V capacitor from TD1 (NS) to V3, pin 6 (NS). Use sleeving. Lay close to terminal strip TD, as shown in Pictorial 3.

( ) Select the two positive (+) leads of the 20-20 μfd 350 V electrolytic capacitor. Connect one lead to TD3 (NS) and the other to TD2 (NS). Connect the negative (-) lead to TE3 (NS).

( ) Connect the positive (+) lead of a 20 μfd 450 V electrolytic capacitor to TE2 (NS). Connect the negative (-) lead to TD1 (NS).

( ) Connect a piece of bare wire from S2 (S-1) to S3 (S-1).

( ) Connect a 1 KΩ resistor (brown-black-red) from TD2 (NS) to TD3 (NS).

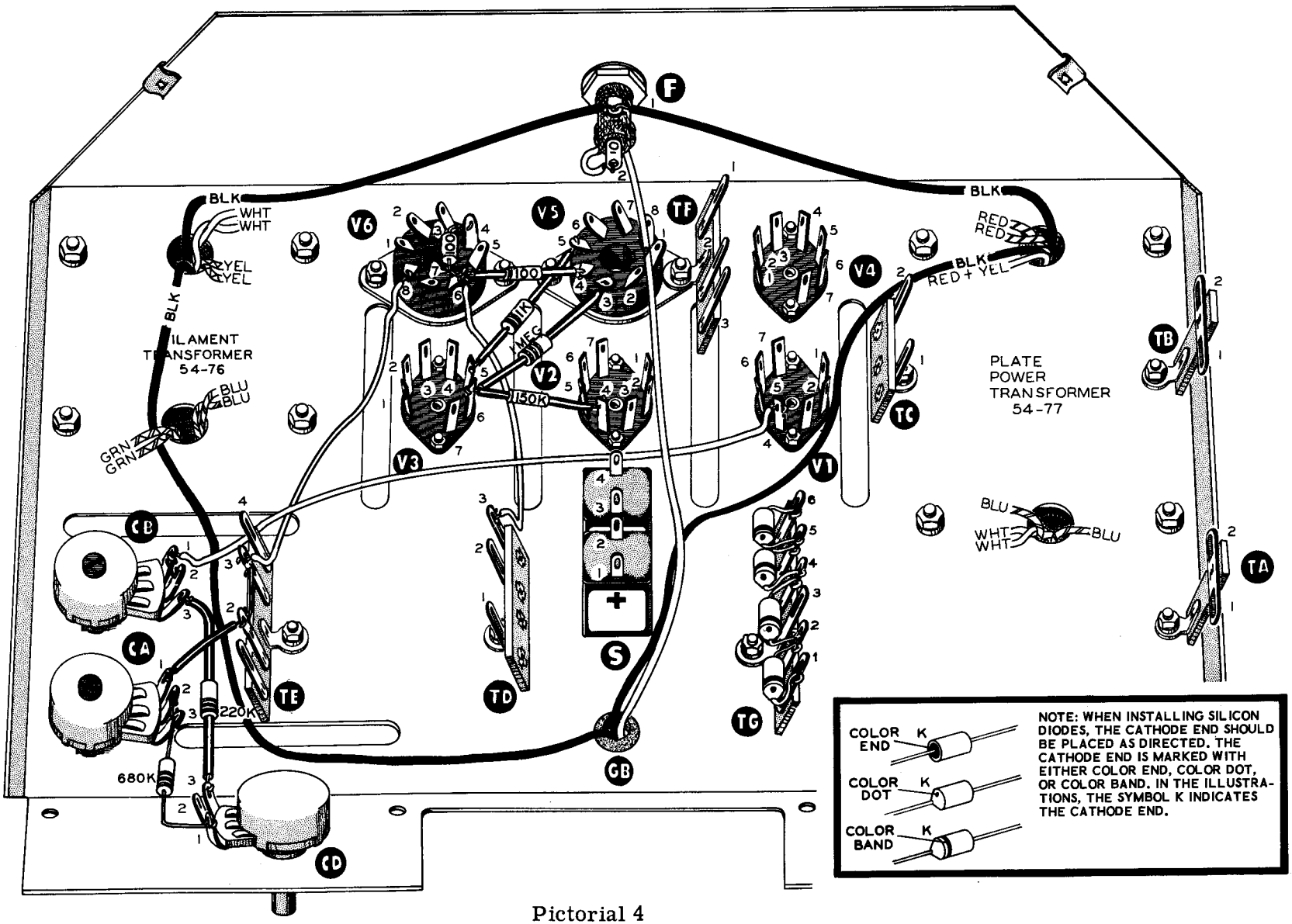
( ) Connect a piece of bare wire from S1 (S-1) to TD2 (S-3).

( ) Connect a .047 μfd 600 V capacitor from V6, pin 8 (NS) to V3, pin 1 (NS). Leave the leads long enough to allow the capacitor body to clear the ventilation hole in the chassis. Use sleeving.

( ) Connect a 1 megohm resistor (brown-black-green) from V3, pin 1 (S-2) to TE4 (NS). Use sleeving.

( ) Connect a 470 KΩ resistor (yellow-violet-yellow) from V6, pin 7 (S-3) to V3, pin 5 (NS).

( ) Connect a 1 KΩ resistor (brown-black-red) from V3, pin 5 (NS) to V6, pin 5 (S-1). Use sleeving.


F
BLK
WHT
WHT
YEL
YEL
BLK
FILAMENT
TRANSFORMER
54-76
BLU
BLU
GRN
GRN
V6
V5
TF
100
.1K
.1MFD
150K
V2
V3
V4
V1
BLK
RED
RED
BLK
RED + YEL
TC
PLATE
POWER
TRANSFORMER
54-77
TB
TA
BLU
WHT
WHT
BLU
CB
CA
TE
220K
680K
CD
S
TD
TG
GB
COLOR END K
COLOR DOT K
COLOR BAND K
NOTE: WHEN INSTALLING SILICON DIODES, THE CATHODE END SHOULD BE PLACED AS DIRECTED. THE CATHODE END IS MARKED WITH EITHER COLOR END, COLOR DOT, OR COLOR BAND. IN THE ILLUSTRATIONS, THE SYMBOL K INDICATES THE CATHODE END.

Pictorial 4

( ) Connect a 1 KΩ resistor (brown-black-red) from V3, pin 5 (S-3) to V5, pin 5 (S-1). Use sleeving. See Pictorial 4.

( ) Connect a 100 Ω resistor (brown-black-brown) from V6, pin 4 (S-1) to V6, pin 6 (NS).

( ) Connect a 100 Ω resistor (brown-black-brown) from V5, pin 4 (S-1) to V6, pin 6 (NS). Use sleeving.

( ) Connect a piece of hookup wire from TD3 (S-3) to V6, pin 6 (S-3).

( ) Connect a 1 megohm resistor (brown-black-green) from V3, pin 6 (NS) to V5, pin 3 (S-3). Use sleeving.

( ) Connect a 150 KΩ resistor (brown-green-yellow) from V3, pin 6 (S-3) to V2, pin 4 (S-3). Use sleeving.

( ) Connect a piece of hookup wire from TE3 (NS) to V6, pin 8 (S-4).

( ) Connect a piece of bare wire from TE2 (NS) to TE3 (NS).

( ) Connect a piece of bare wire from CB2 (S-1) to CB1 (NS).

( ) Connect a piece of bare wire from CA2 (S-1) to CA3 (NS).

( ) Connect a 220 KΩ resistor (red-red-yellow) from CB3 (S-1) to CD3 (S-1). Use sleeving.

( ) Connect a piece of hookup wire from V1, pin 4 (S-1) to CB1 (S-2).

( ) Connect a piece of bare wire from TE2 (NS) to CA1 (S-1). Use sleeving.

( ) Connect a 680 KΩ resistor (blue-gray-yellow) from control CA3 (S-2) to CD1 (S-1).

( ) Connect a silicon diode between lugs 5 (NS) and 6 (S-3) of terminal strip TG. Connect the lead at the marked end to lug 6.

( ) Connect a silicon diode between lugs 3 (NS) and 5 (S-2) of terminal strip TG. Connect the lead at the marked end to lug 5.

( ) Connect a silicon diode between lugs 2 (NS) and 3 (S-3) of terminal strip TG. Connect the lead at the marked end to lug 3.

( ) Connect a silicon diode between lugs 1 (S-5) and 2 (S-2) of terminal strip TG. Connect the lead at the marked end to lug 2.

( ) Mount the fuse holder in location F. Orient as shown in Pictorial 4 and secure with lockwasher and nut supplied with holder. See Figure 3.

( ) Connect the shortest black lead from the power transformer (#54-77) to F1 (NS).

( ) Connect the shortest black lead from the filament transformer (#54-76) to F1 (S-2).

( ) Now set the chassis aside and select the front panel.

( ) Mount binding posts in locations B1, B2, B3, B4, B5, B6 and B7. Mount as shown in Figure 4 and orient the solder lugs as shown in Pictorial 5. Place red binding post caps on B1 and B7 and black caps on the rest. When securing, be sure to keep the holes in the binding post bases vertical.

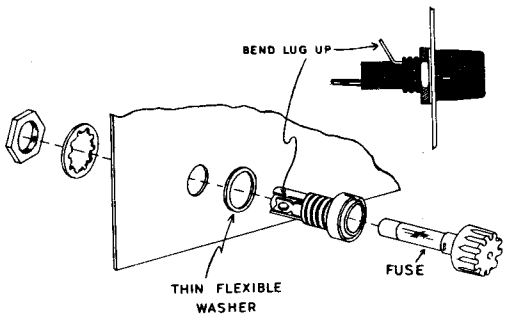


Figure 3

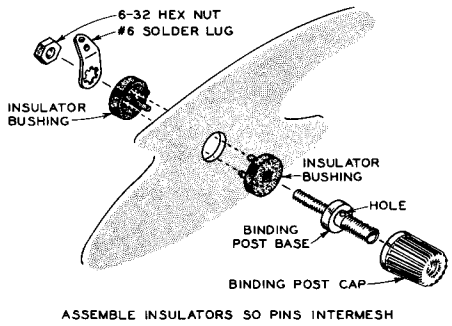
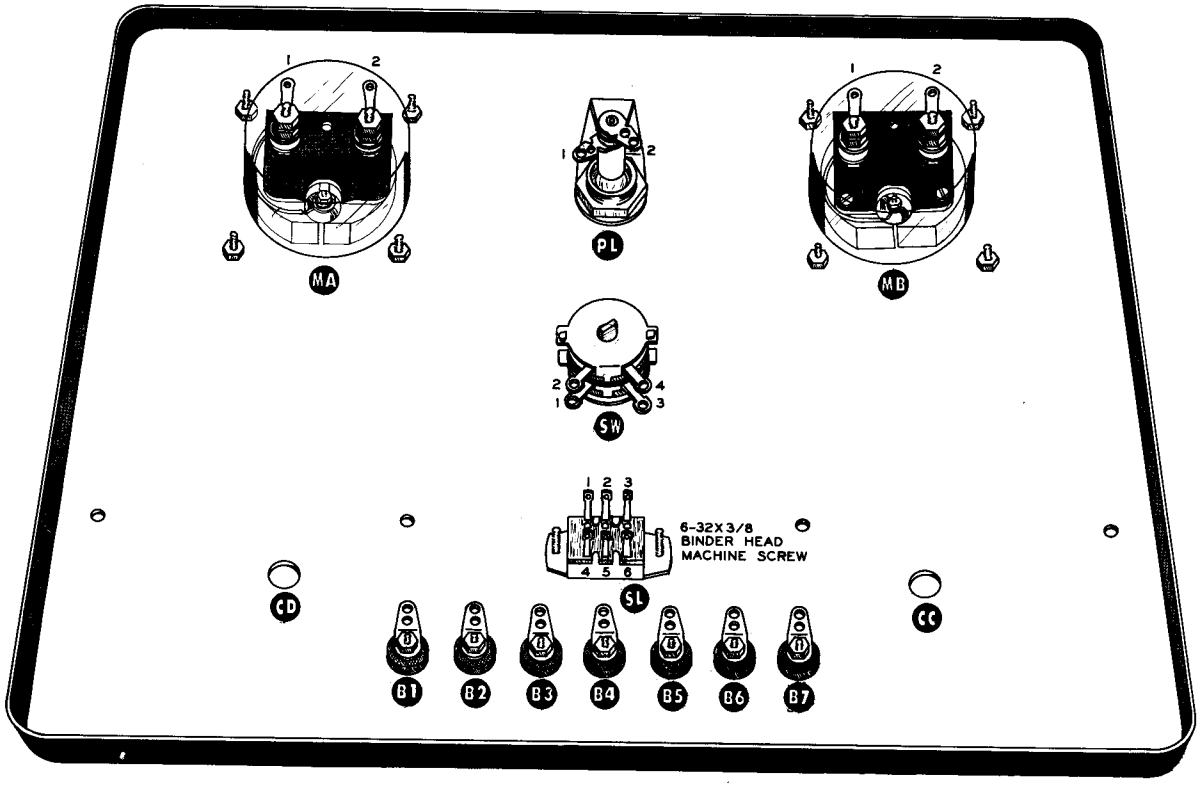
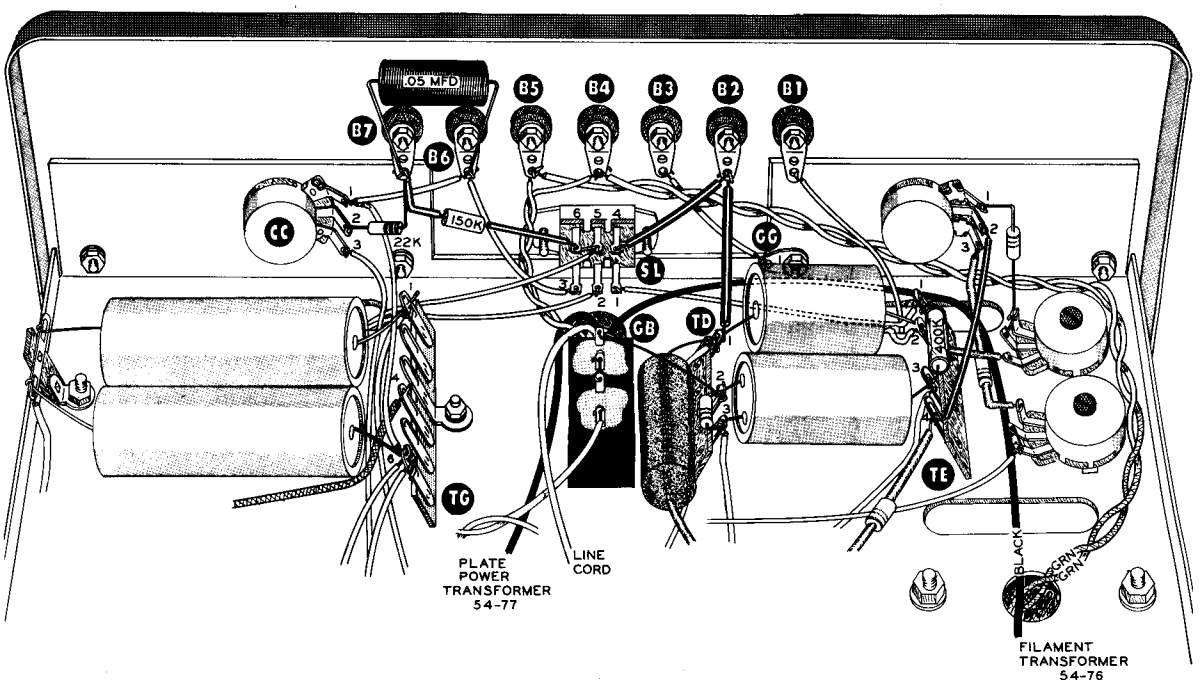


Figure 4



Pictorial 5



Pictorial 6

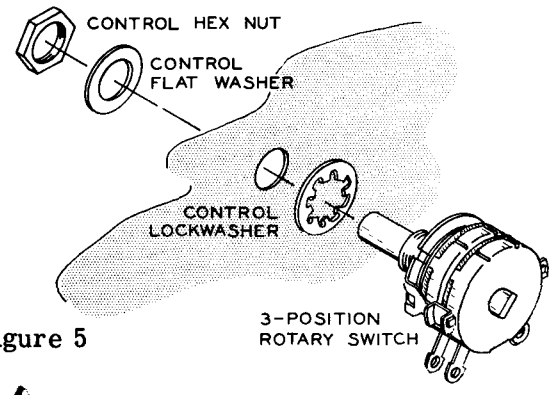


Figure 5

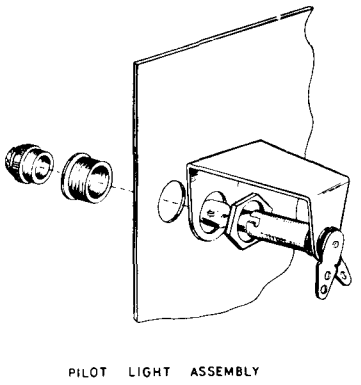


Figure 6

( ) Mount the DPDT slide switch in location SL. Secure with two 6-32 x 3/8" binder head machine screws.

( ) Mount the 3-position rotary switch in location SW. Orient lugs as shown in Pictorial 5 and secure with control lockwasher, flat washer and nut, as shown in Figure 5.

( ) Mount the pilot light assembly in location PL. Orient as shown in Pictorial 5 and secure as shown in Figure 6.

( ) Mount the voltmeter in location MB and the milliammeter in location MA. Secure with lockwashers and nuts supplied with the meters. Use care not to overtighten.

( ) Remove the control nuts and flat washers holding controls CC and CD, and slip the front panel in place. Secure panel by replacing the control flat washers and nuts on the controls. Also place 6-32 x 3/8" binder head machine screws through the four mating front panel and chassis holes. Secure by using a #6 solder lug and nut on the screw in location GG, and #6 lockwashers and nuts on the others. See Pictorial 6.

( ) Connect a piece of hookup wire from B3 (S-1) to solder lug GG (S-1).

( ) Connect a piece of hookup wire from MA2 (S-1) through ventilation slot GC to B1 (S-1). See Pictorials 6 and 7.

( ) Connect a piece of hookup wire from MA1 (S-1) through ventilation slot GC to TE2 (S-4).

( ) Connect a piece of hookup wire from SL1 (S-1) to TE1 (NS).

( ) Connect a 400 KΩ precision resistor from TE1 (S-2) to TE3 (S-4). Avoid excessive heat. Be sure the resistor body does not touch TE2.

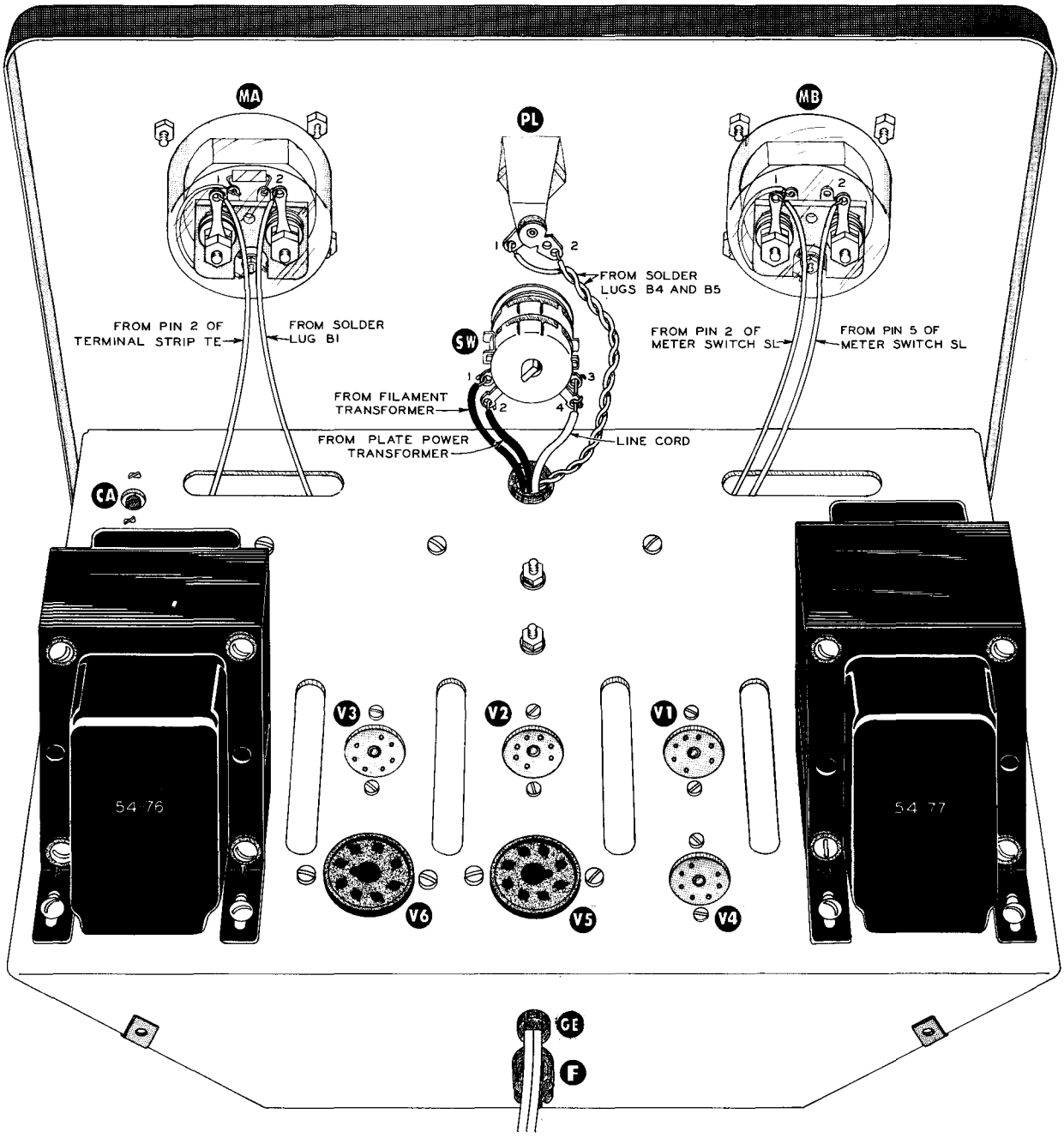
( ) Connect a piece of bare wire from CD2 (S-1) to TE4 (S-2). Use sleeving. Make this lead as short as possible.

( ) Connect a piece of bare wire from SL4 (S-1) to B2 (NS). Use sleeving.

( ) Connect a piece of bare wire from B2 (S-2) to TD1 (S-4). Use sleeving.

( ) Twist together the green leads from the filament transformer. Place along chassis as shown in Pictorial 6 and connect one lead to B4 (NS). Connect the other lead to B5 (NS).

( ) Twist together two 8" lengths of hookup wire and strip the ends of each lead. Select one end of the pair and connect one lead to B4 (S-2) and the other to B5 (S-2). Run the other end of this pair through grommet GB and connect one lead to PL1 (S-1) and the other to PL2 (S-1). Place as shown in Pictorials 6 and 7.


MA
PL
MB
FROM SOLDER LUGS B4 AND B5
FROM PIN 2 OF TERMINAL STRIP TE
FROM SOLDER LUG B1
SW
FROM PIN 2 OF METER SWITCH SL
FROM PIN 5 OF METER SWITCH SL
FROM FILAMENT TRANSFORMER
FROM PLATE POWER TRANSFORMER
LINE CORD
CA
V3
V2
V1
V6
V5
V4
GE
F
54-76
54-77

Pictorial 7

( ) Connect a piece of hookup wire from CC1 (S-2) to B6 (NS). See Pictorial 6.

( ) Connect a piece of hookup wire from B6 (NS) to SL3 (S-1).

( ) Connect a piece of hookup wire from MB1 (S-1) through ventilation slot GA to SL2 (S-1). See Pictorials 6 and 7.

( ) Connect a piece of hookup wire from MB2 (S-1) through ventilation slot GA to SL5 (NS).

( ) Connect a .05 μfd 200 V capacitor from B6 (S-3) to B7 (NS). Position against the front panel as shown in Pictorial 6.

( ) Connect a 22 KΩ resistor (red-red-orange) from CC2 (S-1) to B7 (NS).

( ) Connect a 150 KΩ precision resistor from B7 (S-3) through SL6 (S-1), to SL5 (S-2). Use sleeving.

( ) Place the black lead from the filament transformer (#54-76) through grommet GB and connect it to SW1 (S-1). Orient lead along chassis as shown in Pictorials 4 and 7.

( ) Place the black lead from the power transformer (#54-77) through grommet GB and connect it to SW2 (S-1). Orient lead along chassis as shown in Pictorials 4 and 7.

( ) Connect a piece of bare wire from SW3 (S-1) to SW4 (NS).

( ) Separate the two leads of the line cord for a distance of about 6 1/2" and place them through hole GE. Place one of the leads through grommet GB and connect it to SW4 (S-2). Cut the other lead to proper length and connect it to F2 (S-1). See Pictorials 4 and 7.

( ) Install the line cord bushing on the line cord and insert in hole GE as shown in Figure 7.

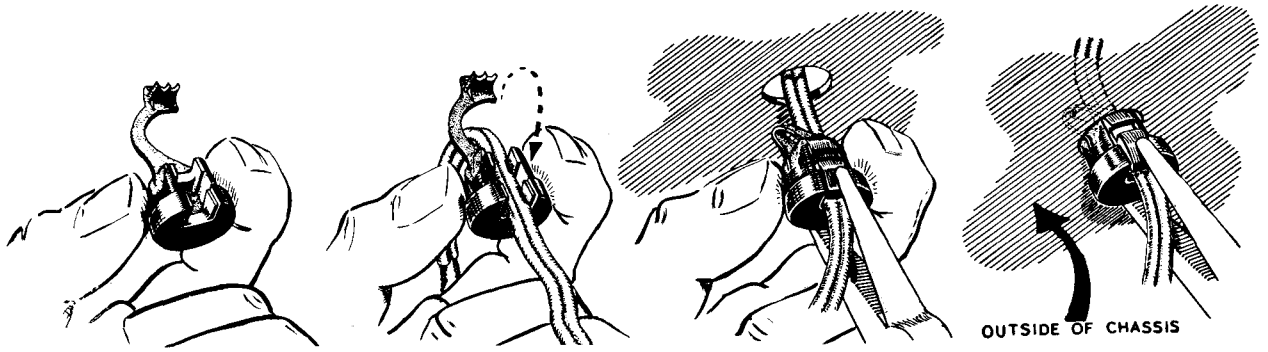


Figure 7

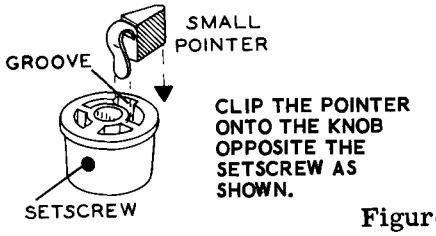


Figure 7A

( ) Install pointers on the three knobs as shown in Figure 7A.

( ) Turn all three controls fully counterclockwise.

( ) Install knobs on controls CC, CD and on switch SW with the pointers at the extreme counterclockwise position.

( ) At this point the wiring is complete. Carefully inspect it to be sure that the connections are all well soldered. Remove any wire cuttings, loose solder, stripped insulation, or other loose material. Check to be sure that you have not accidentally shorted any connections with solder splashes or other foreign bodies.

( ) The rear apron of the chassis has two 3/16" mounting holes. Select the two 6-32 speed-nuts and slip one into place over each hole. Be sure to mount with the smooth surface on the outside of the rear apron. See Figure 8 and Pictorials 4 and 7.

( ) Install the 3 amp fuse in the fuse holder. See Figure 3 on Page 15.

( ) Install the #47 bulb in pilot light holder.



Start SPEED NUT over edge of metal.

Figure 8

( ) Install the tubes in their respective sockets as shown in Figure 9.

( ) Select the cabinet and install the handle, using two #10 x 1/2" screws.

( ) Select the four rubber feet and install them in the bottom of the cabinet as shown in Figure 10.

This completes the assembly of your Regulated Power Supply. Before installing the unit in its cabinet, the following adjustments should be made:

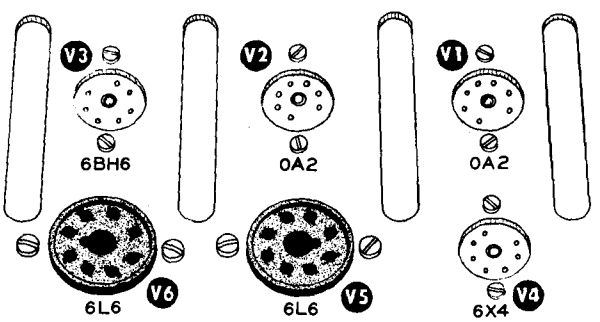


Figure 9



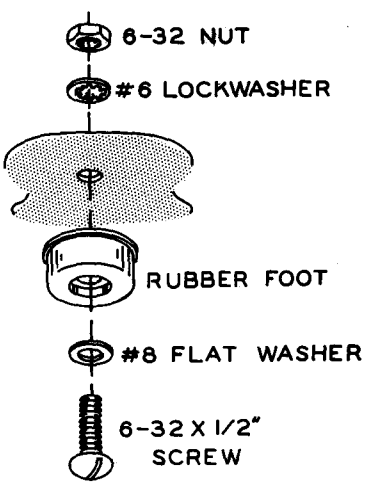


Figure 10

OPERATION AND ADJUSTMENT

( ) Set the power switch (SW) to the OFF position.

( ) Set the bias C- OUTPUT control (CC) and the high voltage B+ OUTPUT control (CD) maximum counterclockwise.

( ) Set the ZERO SET control (CA) maximum clockwise (from top of chassis).

( ) Set the 400 VOLT SET control (CB) maximum counterclockwise (from top of chassis).

( ) Set the meter slide switch (SL) to the B+ OUTPUT position.

NOTE: Do not connect any load to the Supply at this time.

( ) Connect the line cord to a 105-125 volt 50-60 cycle AC outlet.

CAUTION: This instrument will not operate, and may be seriously damaged, if connected to a DC or 25 cycle AC power source or to an AC line of more than 125 volts.

( ) Turn the power switch to STANDBY and allow a few minutes for the tube heaters to come up to operating temperature. Check to be sure that the heaters of the 6X4, 6BH6, and 6L6's are lit.

( ) Turn the power switch to ON. If the unit is functioning properly, the output current will be zero (0) and the voltage will be between zero (0) and 100 volts.

( ) Adjust the ZERO SET control (CA) counterclockwise until the output voltage reads zero (0).

( ) Now set the high voltage B+ OUTPUT control (CD) maximum clockwise.

( ) Adjust the 400 VOLT SET control (CB) clockwise until the output voltage reads 400 volts.

( ) Now set the high voltage B+ OUTPUT control (CD) back to its maximum counterclockwise position.

( ) Because the ZERO SET and 400 VOLT SET controls interact with each other, it will be necessary to repeat the above four steps about 3 or 4 times.

( ) This completes the adjustment of the unit. Unplug the instrument and insert it into the cabinet. Secure in the back with two 6-32 x 3/8" binder head machine screws.

( ) For normal operation, connect a jumper lead (piece of bare wire) from the chassis ground terminal (B3) to the adjacent common terminal (B2).

Regulated B+ Output:

Connect the load between the B+ OUTPUT and the COMMON terminals. Rotate the B+ OUTPUT control clockwise to the desired output voltage. Read the output current and voltage (400 volt scale) on the two meters. Use care not to exceed the output ratings. Note the warning red line at 100 ma on the milliammeter.

Bias C- Output:

Connect the bias or other light load between the C- OUTPUT and COMMON terminals. Set the meter slide switch in the C- OUTPUT position. Rotate the C- OUTPUT control clockwise to the desired output voltage and read the output voltage (150 volt scale) on the voltmeter.

NOTE: The bias output current is not metered.

Filament Output:

Connect the filament load to the two filament voltage output terminals. This is a fixed 6.3 volts, with a maximum output of 4 amperes. Use care not to exceed this output rating.

NOTE: This output is insulated to withstand up to 1500 volts DC between it and common.

OUTPUT METER ACCURACY

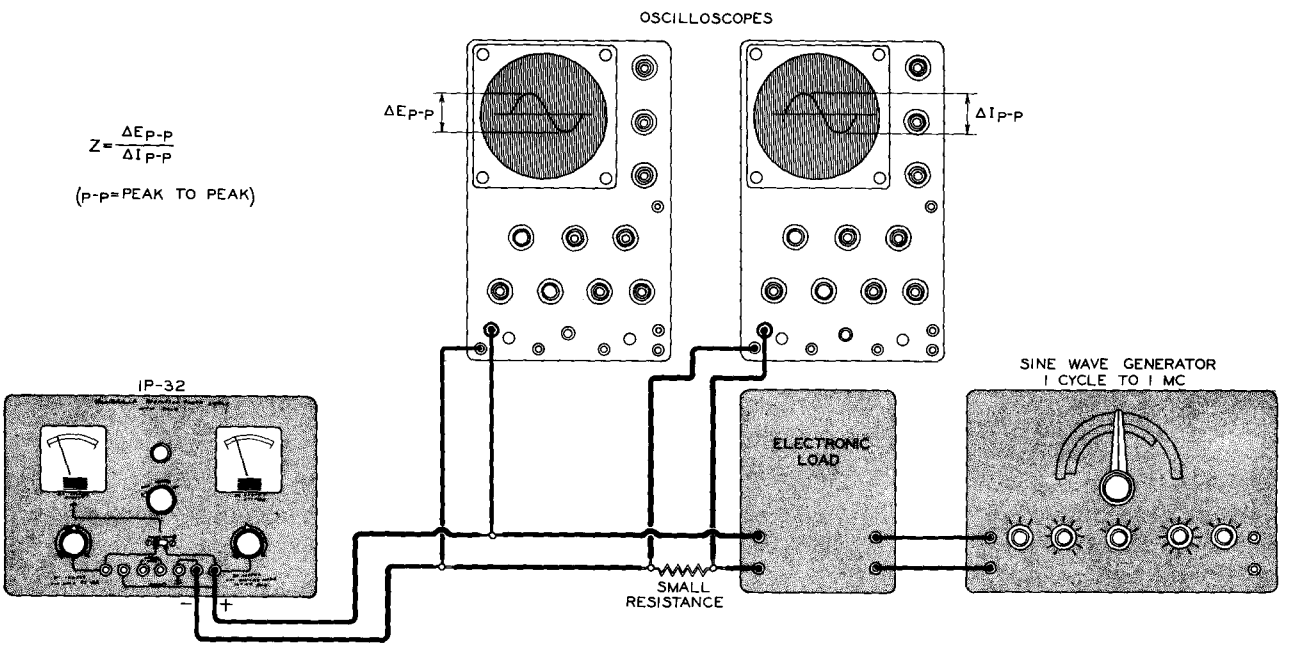
Output Current (full scale)..... ±2%
Output Voltage (full scale)..... ±3%

NOTE: Both output meters have a full scale accuracy of ±2%, but because of the ±1% dropping resistor in series with the voltmeter, its total accuracy is ±3%. However, it is not likely that the meter and resistor tolerance would both be additive maximums. Therefore we would assume its total accuracy to be well within the specified ±3%.

EQUIPMENT LIST

Test equipment used in making the specification measurements for the Heathkit Regulated Power Supply:

Generators:.....	Heathkit Audio Generators
Meters:.....	Hewlett-Packard Model 400D Audio Voltmeter Waveforms Model 520-A Audio Voltmeter Triplett Model 660 AC Wattmeter Heathkit Vacuum Tube Voltmeter Heathkit Multimeter
Oscilloscope:.....	Tektronix 515 Oscilloscope
Line Voltage Control:.....	General Radio Company Type W5MT Variac
Electronic Load:.....	Special design Static and Dynamic Load



DETERMINATION OF OUTPUT IMPEDANCE USING ELECTRONIC STATIC AND DYNAMIC LOAD